Bond Sphere: Diatomic Molecules as S^3 and their Natural Sturmian Basis*

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A diatomic molecule is geometrically a single three-sphere S^3 with two weighted poles—the *bond sphere*—in which the internuclear distance R encodes as a polar angle γ via $\cos\gamma = (p_0^2 - p_R^2)/(p_0^2 + p_R^2)$ and all cross-atom matrix elements are $SO(4)$ Wigner D-matrix elements evaluated at the bond rotation $e^{i\gamma A_y}$. The natural basis for this geometry is a set of Sturmian functions sharing a single momentum parameter p_0, whose β_k eigenvalues are determined by the two-center prolate spheroidal eigenvalue problem, and whose uniform diagonal $\varepsilon_k = -p_0^2/2$ is the hallmark of the Sturmian construction. Every atom-centered approximation to this basis fails for heteronuclear molecules by a precise mechanism: at the shared p_0 set by the deeper well, the kinetic penalty $p_0^2/2$ exceeds the self-nuclear attraction $Z_B p_0$ for the lighter center, and no atom-centered cross-nuclear correction can compensate. The correct cross-nuclear integrals require the molecular Sturmian functions, which are the two-center prolate spheroidal eigenfunctions. Detailed numerical validation, including the complete diagnostic arc (v0.9.11–v0.9.30), is presented in the companion computational paper [4].

I. INTRODUCTION

In the GeoVac framework [1–3], quantum numbers (n, l, m) are not abstract labels but geometric coordinates on a three-sphere S^3. The principal quantum number n indexes latitude shells, each containing n^2 nodes; the orbital angular momentum l and magnetic quantum number m locate a node within its shell. The graph Laplacian over these nodes is mathematically equivalent to the Laplace–Beltrami operator on the unit S^3, and the continuous Schrödinger equation emerges as its flat-space projection via Fock's 1935 stereographic map [3, 5].

This geometric picture is complete for single atoms. For molecules, the current implementation stitches together two separate S^3 manifolds—one per atom—using approximate cross terms: Mulliken overlap for cross-nuclear attraction, Fourier convolution for direct Coulomb integrals, and STO-overlap bridges for inter-center hopping [4]. The result is basis set superposition error (BSSE), approximate integrals, and no natural coordinate system for the bonding region.

This paper proposes the resolution: the *bond sphere*. A diatomic molecule is described by a single S^3 with two weighted poles, one per nucleus. The atomic S^3 manifolds are the $R \to \infty$ limit. Cross-atom matrix elements that are currently approximated become exact $SO(4)$ Wigner D-matrix elements. The natural basis for this geometry is the Sturmian basis, in which all functions share a single momentum parameter p_0 and therefore live on the same S^3 by construction. The β_k eigenvalues that encode nuclear attraction are determined by the two-center prolate spheroidal eigenvalue problem—whose coordinate surfaces are precisely the confocal ellipsoids of the bond sphere metric.

The paper is organized as follows. Sections III and IV establish the single-atom and diatomic bond sphere geometry. Section V gives the Runge–Lenz alignment argument for why the bond rotation is generated by A_y. Section VI treats heteronuclear asymmetry—the egg. Section VII identifies the kinetic membrane as the Fiedler vector zero-crossing. Section VIII introduces the Sturmian basis. Section IX develops the β_k eigenvalue problem via prolate spheroidal separation. Section X proves why atom-centered approximations fail. Section XI states the uniform diagonal theorem. Section XII gives the molecular orbital FCI specification. Section XV concludes.

II. WHAT IS KNOWN VS. WHAT IS NEW

A. Known Results (Prior Work)

- *Shibuya and Wulfman (1965)* [6]: The two-center Coulomb problem in momentum space maps to a single S^3 with two poles. Sturmian basis functions on this sphere provide a complete, orthogonal expansion.
- *Aquilanti and Cavalli (1980s–1990s)* [7, 8]: Hyperspherical harmonic approach to molecular problems, including systematic use of $SO(4)$ Wigner D-matrices and the relationship between internuclear distance and angular pole separation on S^3.
- *Avery (2000s)* [10]: Extended the Sturmian basis to general molecular systems with practical computational methods.
- $SO(4)$ *representation theory* [11, 12]: The Wigner D-matrix elements of $SO(4)$ in the (n, l, m) basis, generated by angular momentum $\mathbf{L}$ and the Runge–Lenz vector $\mathbf{A}$.

* Paper 8–9 in the Geometric Vacuum series

- *Sturmian basis functions*: Established by Shull and Löwdin [13], Rotenberg [14], and Aquilanti–Avery [15, 16]. Orthogonality under $1/r$ weight and completeness for the Coulomb problem.

B. New Contributions

1. **Bond sphere as a GeoVac lattice object.** The bond sphere is the natural molecular generalization of the single-atom S^3 lattice (Papers 0, 1, 7), with within-block edges generated by $T_\pm, L_\pm$ and cross-block edges weighted by D-matrix elements at $g(\gamma(R))$.
2. **Runge–Lenz alignment argument** (Sec. V). The poles are forced onto the bond axis by dynamics: the Runge–Lenz vector of each atom orients toward the other nucleus.
3. **Heteronuclear egg geometry** (Sec. VI). For $Z_A \neq Z_B$, the bond sphere metric is a prolate spheroid with the heavier nucleus at the fat end.
4. **Kinetic membrane** (Sec. VII). The Fiedler vector zero-crossing identifies a discrete, combinatorial analog of Bader's bond critical point.
5. **Identification that the bond sphere requires the Sturmian basis** (Sec. VIII). The single-S^3 geometry is only consistent when all basis functions share one p_0.
6. **Proof that atom-centered Sturmian bases fail for heteronuclear molecules** (Sec. X), with the precise threshold $\beta > p_0/(2Z_B)$ for binding.
7. **Uniform diagonal theorem** (Sec. XI). In the molecular Sturmian basis, $\varepsilon_k = -p_0^2/2$ for every orbital.
8. **Molecular orbital FCI specification** (Sec. XII). A complete theoretical specification for FCI in the molecular Sturmian basis.

III. THE SINGLE-ATOM CASE AS A SPECIAL LIMIT

The existing GeoVac single-atom lattice is the $\gamma = 0$ limit of the bond sphere, so all results of Papers 0–7 are preserved.

A. The $T_\pm, L_\pm$ Operators as $SO(4)$ Generators

The GeoVac paraboloid lattice is generated by four transition operators (Paper 7, Eqs. (5)–(6)):

$$L_\pm|n,l,m\rangle \propto |n,l,m\pm 1\rangle, \tag{1}$$

$$T_\pm|n,l,m\rangle \propto |n\pm 1,l,m\rangle. \tag{2}$$

The angular momentum operators L_x, L_y, L_z satisfy $\mathfrak{su}(2)$:

$$[L_i, L_j] = i\epsilon_{ijk}L_k. \tag{3}$$

The scaled Runge–Lenz vector $\mathbf{A}$, generated by $T_\pm$ within each n-shell, satisfies [11]:

$$[L_i, A_j] = i\epsilon_{ijk}A_k, \tag{4}$$

$$[A_i, A_j] = i\epsilon_{ijk}L_k. \tag{5}$$

The $\mathfrak{so}(4) \cong \mathfrak{su}(2) \oplus \mathfrak{su}(2)$ structure is exhibited by

$$\mathbf{J}^{(\pm)} = \tfrac{1}{2}(\mathbf{L} \pm \mathbf{A}), \tag{6}$$

which satisfy two independent, commuting $\mathfrak{su}(2)$ algebras. For principal quantum number n, both carry spin $j = (n-1)/2$, giving the $(n-1)/2 \otimes (n-1)/2$ representation with Casimir

$$\mathbf{L}^2 + \mathbf{A}^2 = n^2 - 1, \tag{7}$$

recovering the S^3 eigenvalue spectrum $\lambda_n = -(n^2-1)$ of Paper 7.

B. Covariance and the Single-Atom Limit

Because $T_\pm, L_\pm$ generate $SO(4)$, the graph Laplacian $L = D - A$ is covariant under $SO(4)$ transformations. The second atom can be introduced as an $SO(4)$-rotated copy, with the rotation angle determined by R. Cross-atom matrix elements are then $SO(4)$ Wigner D-matrix elements.

At $\gamma = 0$ (coincident poles): $g(0) = e$, the D-matrix reduces to Kronecker deltas, cross-block edges vanish, and the single-center lattice of Papers 0–7 is recovered. The 18 symbolic proofs, the dimensionless vacuum principle, and the universal constant $-1/16$ are unchanged.

IV. THE BOND SPHERE CONSTRUCTION

A. The Angle–Distance Relationship

The internuclear distance R maps to a polar angle γ between two poles on S^3. The momentum conjugate to R defines

$$p_R = \frac{1}{R}, \tag{8}$$

and the angular separation is

$$\cos\gamma = \frac{p_0^2 - p_R^2}{p_0^2 + p_R^2}, \tag{9}$$

which is the fourth component n_4 of the Fock projection evaluated at $|\mathbf{p}| = p_R$. The sine follows as

$$\sin\gamma = \frac{2p_0\, p_R}{p_0^2 + p_R^2} = \Omega(p_R)\cdot p_R, \tag{10}$$

where Ω is the Fock conformal factor.

a. Limiting behavior. At $R \to \infty$: $\gamma \to 0$ (coincident poles, single-atom limit). At $R \to 0$: $\gamma \to \pi$ (antipodal poles, united-atom limit).

B. Eigenfunctions as $SO(4)$ Wigner D-Matrix Elements

When a second pole is introduced at angle γ, the cross-center matrix elements are

$$\langle n', l', m' | U(g(\gamma)) | n, l, m \rangle = D^{(n)}_{(l'm'),(lm)}(g(\gamma)), \quad (11)$$

where $g(\gamma)$ is the $SO(4)$ rotation that maps one pole to the other. The current Mulliken/Fourier/bridge cross-atom terms are approximations to these exact D-matrix elements.

C. The Bond-Sphere Lattice

a. Nodes. Each node is labeled (n, l, m, α) with $\alpha \in \{A, B\}$. Total spatial orbitals: $2N_{\text{sp}}$.

b. Within-block edges. Generated by $T_\pm, L_\pm$ at each center, as in the single-atom lattice.

c. Cross-block edges. Weighted by D-matrix elements:

$$w_{(n'l'm',B),(nlm,A)} = D^{(n)}_{(l'm'),(lm)}(g(\gamma(R))). \quad (12)$$

d. Hamiltonian structure.

$$H_1 = \begin{pmatrix} H_A & V_{AB} \\ V_{AB}^\dagger & H_B \end{pmatrix}, \quad (13)$$

where H_A, H_B are single-atom Hamiltonians and V_{AB} has elements given by Eq. (12) modulated by nuclear charges and kinetic scale.

D. The Opposite-Sign Rotation

The bond rotation is not a spatial rotation. Since $\mathbf{A} = \mathbf{J}^{(+)} - \mathbf{J}^{(-)}$, the rotation generated by A_y acts as

$$e^{i\gamma A_y} = e^{i\gamma J_y^{(+)}} \, e^{-i\gamma J_y^{(-)}}, \quad (14)$$

where the $\mathbf{J}^{(+)}$ and $\mathbf{J}^{(-)}$ sectors rotate in opposite directions: $+\gamma$ and $-\gamma$. A pure spatial rotation $e^{i\alpha L_y}$ gives same-sign rotation $d^j(+\alpha) \otimes d^j(+\alpha)$, preserving radial structure. The bond rotation gives $d^j(+\gamma) \otimes d^j(-\gamma)$, mixing $s, p, d, \ldots$ orbitals within each n-shell. This s–p hybridization is precisely what chemical bonding requires.

The D-matrix in the (n, l, m) basis is

$$D^{(n)}_{(l'm'),(lm)}(\gamma) = \sum_{m^+,m^-} \sum_{m'^+,m'^-} C^n_{l'm';m'^+m'^-} \, C^n_{lm;m^+m^-} \times d^j_{m'^+,m^+}(+\gamma) \, d^j_{m'^-,m^-}(-\gamma), \quad (15)$$

where $j = (n-1)/2$ and $C^n_{lm;m^+m^-}$ are the $SO(4)$ Clebsch–Gordan coefficients coupling $|j, m^+\rangle \otimes |j, m^-\rangle \to |l, m\rangle$ with $m = m^+ + m^-$.

E. The Antipodal Limit

At $\gamma = \pi$ ($R = 0$, united atom):

$$D^{(n)}(\pi)^2 = \mathbf{I}_{n^2}, \quad (16)$$

with eigenvalues ± 1 and parity structure $|l, m\rangle \to (-1)^{n-1+l+|m|} \, |l, -m\rangle$—the Runge–Lenz parity, not spatial parity $(-1)^l$.

V. THE RUNGE–LENZ ALIGNMENT ARGUMENT

The bond rotation $e^{i\gamma A_y}$ is generated by A_y, a component of the Runge–Lenz vector perpendicular to the bond axis. This choice is not a convention but a dynamical consequence.

For a hydrogen atom, the Runge–Lenz vector $\mathbf{A}$ points from the nucleus toward the perihelion of the classical orbit. In a diatomic molecule, the electrostatic field of the partner nucleus breaks the spherical symmetry of each atom's potential. The lowest-energy configuration aligns the Keplerian orbit so that its perihelion points toward the attracting partner—that is, $\mathbf{A}$ orients along the internuclear axis. With $\mathbf{A}$ aligned along $\hat{z}$ (the bond axis), the rotation that maps one pole to the other is generated by A_y (or A_x), which is perpendicular to the bond.

This is the physical origin of s–p hybridization in the bond sphere picture: the Runge–Lenz vector aligns with the bond, and rotation about a perpendicular axis mixes orbital angular momentum states. The hybridization is not an ansatz imposed on the basis—it is a consequence of the two-center electrostatic potential acting on the dynamical symmetry generators of each atom.

VI. HETERONUCLEAR GEOMETRY: THE EGG

For a homonuclear diatomic ($Z_A = Z_B$), the bond sphere has uniform metric and full $SO(4)$ symmetry. For a heteronuclear diatomic ($Z_A \neq Z_B$), the geometry changes qualitatively.

A. Metric Distortion

The Fock conformal factor $\Omega = 2p_0/(p^2 + p_0^2)$ maps momentum space to S^3. When the two nuclei have unequal charges, the natural momentum scale differs between the two poles: center A (charge Z_A) favors $p_0 \sim Z_A$, while center B (charge Z_B) favors $p_0 \sim Z_B$. With a shared p_0 (as required by the single-S^3 construction), the conformal factor is distorted asymmetrically about the two poles.

The result is a bond sphere whose intrinsic metric is not uniform. The heavier nucleus ($Z_A > Z_B$) occupies a

larger solid angle on the sphere—a "fat end"—while the lighter nucleus occupies a smaller solid angle—a "narrow end." The overall shape is a prolate spheroid, with the heavier nucleus at the fat end and the lighter nucleus at the narrow end. We call this the *egg*.

B. Bond Polarity as Egg Asymmetry

The degree of asymmetry is set by the ratio Z_A/Z_B. For LiH ($Z_A/Z_B = 3$), the egg is noticeably asymmetric. For H_2 ($Z_A/Z_B = 1$), the egg is a round sphere. This asymmetry is the geometric encoding of bond polarity: a polar bond is an asymmetric egg, and a nonpolar bond is a symmetric sphere.

C. Topology vs. Metric

The node count per shell (n^2 nodes at latitude n from each pole) is fixed by the Paper 0 packing topology. The egg deformation changes the *node weights*—the effective nuclear attraction felt by each node—but not the *graph topology*. The adjacency structure ($T_\pm, L_\pm$ edges) is unchanged; only the diagonal elements of the Hamiltonian (the node weights) and the cross-block edge weights (the D-matrix elements) are affected by the asymmetry.

D. Symmetry Breaking: $SO(4) \to SO(2)$

The unequal pole weights break $SO(4)$ to the axial $SO(2)$ subgroup preserving the bond axis. The magnetic quantum number m remains a good quantum number (generating $\sigma, \pi, \delta, \ldots$ labels), while the off-axis generators $L_\pm$ are no longer symmetries:

$$[\Delta_{\text{asym}}, L_z] = 0, \tag{17}$$

$$[\Delta_{\text{asym}}, L_\pm] \neq 0 \quad \text{for } Z_A \neq Z_B. \tag{18}$$

E. Prolate Spheroidal Coordinates

The natural coordinate system for the egg is prolate spheroidal coordinates (ξ, η, φ) with foci at the two nuclear poles. The coordinate surfaces are confocal ellipsoids ($\xi = \text{const}$) and hyperboloids ($\eta = \text{const}$). These are precisely the level sets of the distorted bond sphere metric: the ellipsoids trace curves of constant "distance from both poles simultaneously," while the hyperboloids trace curves of constant "distance ratio between the two poles." This coordinate system appears again in Sec. IX as the natural domain for the β_k eigenvalue problem.

VII. THE KINETIC MEMBRANE

Between the two poles of the bond sphere lies a region where the edge weights from the two centers are comparable in magnitude. This is the *kinetic membrane*—not a separate geometric object, but an emergent feature of the edge weight pattern.

A. Identification via the Fiedler Vector

In the molecular graph, the kinetic membrane is identified by the *Fiedler vector*—the eigenvector corresponding to the second smallest eigenvalue of the graph Laplacian. The Fiedler vector partitions the graph into two regions (positive and negative components), and the membrane is where the Fiedler vector crosses zero.

For a homonuclear diatomic, the Fiedler vector zero-crossing lies at the midpoint between the two poles, by symmetry. For a heteronuclear molecule, the zero-crossing shifts toward the lighter nucleus. In LiH, the heavier Li pole ($Z = 3$) dominates a larger volume of the bond sphere, pushing the membrane toward the H end.

B. Connection to Bader's Bond Critical Point

The Fiedler vector zero-crossing is the discrete, combinatorial analog of Bader's bond critical point [17] in the Quantum Theory of Atoms in Molecules (QTAIM). Bader's bond critical point is defined as the point where $\nabla\rho = 0$ along the bond path, where ρ is the electron density. The GeoVac graph gives the same object without invoking the electron density: the membrane position is determined purely by the graph Laplacian spectrum.

For heteronuclear molecules, both the Bader critical point and the Fiedler zero-crossing shift toward the electropositive (lighter) atom. The membrane is thinner (steeper Fiedler gradient) for strongly polar bonds and thicker (shallower gradient) for nonpolar bonds.

VIII. THE STURMIAN BASIS

The bond sphere hypothesis requires all basis functions to live on a single S^3 parameterized by one momentum scale p_0. The hydrogenic eigenstate basis uses $p_0(n) = Z/n$—a different sphere per shell—which is geometrically inconsistent with the single-S^3 construction. The Sturmian basis resolves this.

A. Definition

The Sturmian basis functions S_{nlm} are solutions to the scaled Coulomb problem:

$$\left(-\tfrac{1}{2}\nabla^2 - \frac{\beta_n Z}{r}\right) S_{nlm} = E_0\, S_{nlm}, \tag{19}$$

where $E_0 = -p_0^2/2$ is a fixed reference energy shared by all functions, and

$$\beta_n = \frac{p_0\, n}{Z} \tag{20}$$

is the shell-dependent scaling that enforces this.

The key property: all Sturmian functions share the same E_0 regardless of n. Under Fock's projection, all functions live on the *same* S^3 of radius p_0.

B. Verification

Substituting effective charge $\tilde{Z} = \beta_n Z$ into the hydrogenic eigenvalue:

$$E_0 = -\frac{(\beta_n Z)^2}{2n^2} = -\frac{(p_0 n/Z)^2 Z^2}{2n^2} = -\frac{p_0^2}{2}, \tag{21}$$

independent of n, as required.

C. Orthogonality under $1/r$ Weight

Sturmian functions satisfy

$$\left\langle S_{n'l'm'} \middle| \frac{1}{r} \middle| S_{nlm} \right\rangle = \frac{p_0}{n}\, \delta_{nn'}\, \delta_{ll'}\, \delta_{mm'}. \tag{22}$$

The $1/r$-weighted completeness precisely matches the Coulomb kernel in cross-center integrals, making the two-center problem exact.

D. Geometric Inconsistency of the Hydrogenic Basis

The hydrogenic eigenstate ψ_{nlm} has $p_0(n) = Z/n$. For the cross-center matrix element $\langle \psi^A_{n'l'm'} | -Z_B/r_B | \psi^A_{nlm} \rangle$ with $n' \neq n$, the bra lives on S^3 with radius Z/n' and the ket on S^3 with radius Z/n. The integral involves two different Fock projections—a convolution over mismatched spheres. This is the origin of the $\sin\gamma$ form factor in the current SW module: a lowest-order correction for this geometric mismatch.

In the Sturmian basis, the mismatch is zero by construction.

E. SW Integrals Are Exact in the Sturmian Basis

This is the central result. In the Sturmian basis:

$$\boxed{\left\langle S^A_{n'l'm'} \middle| -\frac{Z_B}{r_B} \middle| S^A_{nlm} \right\rangle = -\frac{Z_B}{p_0}\, D^{(n)}_{(l'm'),(lm)}(\gamma),} \tag{23}$$

exactly, with no form factor.

The derivation: the cross-center Coulomb potential is related to the same-center potential by an $SO(4)$ rotation (Eq. (11)). Using Sturmian orthogonality (Eq. (22)), the $1/r_A$ matrix element collapses to a Kronecker delta, and the sum over intermediate states reduces to a single D-matrix element.

When the same calculation is attempted with hydrogenic eigenstates of different p_0 values, the orthogonality does not apply, and one obtains the mismatch integral that, expanded to leading order, gives the $\sin\gamma$ form factor.

F. The Self-Consistency Condition

The momentum parameter p_0 is determined by

$$p_0 = \sqrt{-2E_{\text{mol}}}, \tag{24}$$

where E_{mol} is the ground-state eigenvalue of $H(p_0)$. This is a self-consistent fixed point, whose existence follows from the intermediate value theorem: $f(p_0) = \sqrt{-2E_{\text{mol}}(p_0)}$ maps $\mathbb{R}^+$ to $\mathbb{R}^+$, with $f(p_0) > p_0$ for small p_0 (diffuse basis, weakly negative energy) and $f(p_0) < p_0$ for large p_0 (compact basis, bounded energy).

G. γ Encodes Both Geometry and Energy

At self-consistency:

$$\gamma = \arccos\left(\frac{p_0^2 - p_R^2}{p_0^2 + p_R^2}\right), \tag{25}$$

with $p_0 = \sqrt{-2E_{\text{mol}}}$. Both derivatives $\partial\gamma/\partial R \neq 0$ and $\partial\gamma/\partial E_{\text{mol}} \neq 0$, confirming that γ carries information about both bond length and molecular energy simultaneously.

H. Backward Compatibility with Papers 0–7

In the limit $p_0 \to Z/n$ per shell: $\beta_n \to 1$, the Sturmian equation becomes the ordinary hydrogenic equation, and the atomic eigenvalue $-Z^2/(2n^2)$ is recovered. The self-consistency loop has no single-atom analog because $E_n = -Z^2/(2n^2)$ already determines $p_0(n) = Z/n$. All single-atom results are the n-resolved limit of the Sturmian construction.

IX. THE β_k EIGENVALUE PROBLEM

For a single-center Sturmian, $\beta_n = p_0 n/Z$ is determined by the nuclear charge alone. For a two-center molecular Sturmian, β_k is the eigenvalue of the prolate spheroidal equation with foci at the two nuclei.

A. Prolate Spheroidal Separation

The Sturmian equation with the two-center molecular potential $V = -Z_A/r_A - Z_B/r_B$ separates in prolate spheroidal coordinates (ξ, η, φ) with foci at the nuclear positions. The separation yields an angular eigenvalue problem (in η) and a radial eigenvalue problem (in ξ), coupled by a separation constant A.

The angular equation at fixed magnetic quantum number m and spheroidal parameter $c = p_0 R/2$ is

$$\frac{d}{d\eta}\left[(1-\eta^2)\frac{dS}{d\eta}\right] + \left(A_{m,n_{\rm sph}} - c^2\eta^2 - \frac{m^2}{1-\eta^2}\right) S = 0, \tag{26}$$

whose eigenvalues $A_{m,n_{\rm sph}}(c,b)$ depend on c and the charge asymmetry parameter $b = (Z_A - Z_B)R/2$.

The radial equation is

$$\frac{d}{d\xi}\left[(\xi^2-1)\frac{dR}{d\xi}\right] + \left(-A + c^2\xi^2 + 2a\xi - \frac{m^2}{\xi^2-1}\right) R = 0, \tag{27}$$

where $a = \beta_k(Z_A + Z_B)R/2$ encodes the molecular Sturmian eigenvalue β_k.

B. The Matching Condition

The molecular β_k for each orbital $(m, n_{\rm sph}, n_{\rm rad})$ is determined by the matching condition: the angular separation constant $A_{m,n_{\rm sph}}(c,b)$ and the radial eigenvalue must be consistent. Specifically, β_k is found by root-finding (brentq) on the condition

$$A_{\rm ang}(c,b) + L_{\rm rad}(c, a(\beta_k)) = 0, \tag{28}$$

where $L_{\rm rad}$ is the top radial eigenvalue at the given parameters.

C. Key Result

Theorem 1 (Molecular Sturmian β_k for LiH)**.** *For LiH at $R = 3.015$ bohr with $p_0 = \sqrt{-2E_{mol}}$, the molecular Sturmian β_k values computed via prolate spheroidal separation are:*

Orbital	β_k	*Character*
1σ	*1.033*	*Li 1s core*
2σ	*1.950*	*Li-H bonding*
1π	*1.976*	*Li-H π*

All $n = 3$ orbitals have $\beta_k > 2.3$. The H_2^+ ground-state energy computed by this method is $E = -1.096$ Ha (exact: -1.103 Ha, 0.6% error).

D. Why Prolate Spheroidal Coordinates Are Natural

The separability of the Sturmian equation in prolate spheroidal coordinates is not a mathematical accident. The coordinate foci are the two nuclear poles, and the coordinate surfaces are the confocal ellipsoids that are the level sets of the distorted bond sphere metric (Sec. VI). The coordinates respect both poles simultaneously—which is precisely the requirement for a basis that lives on the bond sphere rather than on a single-center sphere.

X. WHY ATOM-CENTERED BASES FAIL FOR HETERONUCLEAR MOLECULES

Theorem 2 (Atom-centered Sturmian failure)**.** *For a heteronuclear diatomic with $Z_A/Z_B = k > 1$, there is no atom-centered Sturmian basis with shared p_0 that simultaneously binds both centers.*

Proof sketch. The shared p_0 is determined by the deeper well (Z_A). At this p_0, the one-electron diagonal for the lightest orbital of center B is

$$\varepsilon_1(p_0) = \frac{p_0^2}{2} - Z_B\, p_0. \tag{29}$$

This is positive (unbound) when $p_0 > 2Z_B$. For the self-consistent $p_0 \approx \sqrt{-2E_A}$ of the heavier atom with $Z_A \gg Z_B$, this condition is generically satisfied.

The cross-nuclear correction from Z_A on the B-center $1s$ function, evaluated in the atom-centered Sturmian basis, is $\Delta\varepsilon = -(Z_A/p_0)D^{(1)}(\gamma)$. For $n = 1$, $D^{(1)} = 1$, so $\Delta\varepsilon = -Z_A/p_0$. The total diagonal becomes

$$\varepsilon_1^{\rm tot} = \frac{p_0^2}{2} - Z_B p_0 - \frac{Z_A}{p_0}. \tag{30}$$

For binding: $\varepsilon_1^{\rm tot} < 0$, i.e., $\beta_{\rm eff} > p_0/(2Z_B)$. But the atom-centered $\beta_1 = p_0/Z_B$ gives $\varepsilon_1 = p_0^2/2 - Z_B p_0$ plus the cross-nuclear $-Z_A/p_0$. At $p_0 = 3.168$ (LiH): $\varepsilon_1^{\rm tot} = 5.02 - 3.17 - 0.95 = +0.85$ Ha—still positive.

The cross-nuclear correction is insufficient because the atom-centered function has the wrong spatial extent: it was defined against Z_B alone and is too compact near the B center to capture the full A-B cross-nuclear attraction. The correct integral requires the *molecular* Sturmian function—the two-center prolate spheroidal eigenfunction—which has the correct spatial extent near both centers. □

Corollary 1. *For LiH ($Z_A/Z_B = 3$): the binding threshold is $\beta > p_0/(2Z_B) = 1.581$. The 2σ molecular Sturmian achieves $\beta = 1.950$ (Theorem 1), but no atom-centered approximation reaches this threshold.*

This argument is independent of the number of electrons and applies to any heteronuclear diatomic with Z_A/Z_B sufficiently different from 1. The diagnostic arc v0.9.20–v0.9.30 (documented in the companion paper [4]) exhaustively verifies this theorem computationally: every atom-centered variant fails for LiH by the mechanism described above.

XI. THE UNIFORM DIAGONAL THEOREM

Theorem 3 (Uniform diagonal)**.** *In the molecular Sturmian basis with shared p_0, the diagonal of the one-electron Hamiltonian is*

$$\varepsilon_k = -\frac{p_0^2}{2} \tag{31}$$

for every orbital k, regardless of quantum numbers.

Proof. Each molecular Sturmian S_k satisfies $(-\frac{1}{2}\nabla^2 - \beta_k V_{\rm mol})S_k = E_0 S_k$ with $E_0 = -p_0^2/2$. The diagonal matrix element of the full one-electron Hamiltonian $\hat{h}_1 = -\frac{1}{2}\nabla^2 + V_{\rm mol}$ is

$$\langle S_k|\hat{h}_1|S_k\rangle = E_0 + (1-\beta_k^{-1})\,\langle S_k|V_{\rm mol}|S_k\rangle. \tag{32}$$

In the exact molecular Sturmian basis with proper normalization, the $V_{\rm mol}$ expectation value is $-p_0^2/\beta_k$, giving

$$\varepsilon_k = -\frac{p_0^2}{2} - \frac{p_0^2}{\beta_k} + \frac{p_0^2}{\beta_k} = -\frac{p_0^2}{2}. \tag{33}$$

□

a. Physical interpretation. The uniform diagonal means that no orbital has a different diagonal energy. The entire information content of the nuclear attraction is in the *off-diagonal* β_k structure and the D-matrix coupling. The asymmetry between the two centers enters only through the β_k eigenvalues (which differ for σ vs. π MOs and depend on Z_A, Z_B, R) and through the D-matrix rotation angle γ.

b. Consistency with the β_k discussion. The β_k eigenvalues (Sec. IX) parametrize the nuclear attraction: orbital k has effective charge $\beta_k Z$ in the Sturmian equation. This information enters the FCI Hamiltonian through the off-diagonal matrix elements (the D-matrix coupling weighted by β_k-dependent factors), not through the diagonal. The diagonal is uniformly $-p_0^2/2$, and all orbital differentiation is off-diagonal. This is the correct architecture for a bond sphere FCI: the basis is adapted to the molecular potential, and the nuclear charges appear in the coupling structure, not in the orbital energies.

XII. MOLECULAR ORBITAL FCI SPECIFICATION

This section provides a complete theoretical specification for FCI in the molecular Sturmian basis. A reader should be able to implement the v0.9.31 solver from this section alone.

A. Basis Functions

The FCI basis consists of molecular Sturmian functions labeled by $(m, n_{\rm sph}, n_{\rm rad})$—the magnetic quantum number, angular nodal index, and radial nodal index in the prolate spheroidal coordinate system. For each $(m, n_{\rm sph}, n_{\rm rad})$, the β_k eigenvalue is computed by solving the prolate spheroidal matching condition (Eq. (28)) at the self-consistent p_0.

B. Hamiltonian Matrix Elements

a. Diagonal. Every orbital has the same diagonal energy:

$$\langle k|\hat{h}_1|k\rangle = -\frac{p_0^2}{2}. \tag{34}$$

Nuclear repulsion $V_{NN} = Z_A Z_B/R$ is added as a constant.

b. One-electron off-diagonal. The off-diagonal H_1 matrix elements couple MOs with the same m at the bond angle γ. These are the D-matrix elements weighted by the nuclear charges:

$$\langle k'|\hat{h}_1|k\rangle = -\frac{Z_A+Z_B}{p_0}\,D^{(n)}_{k',k}(\gamma)\,\beta_k, \quad k' \neq k,\ m_{k'} = m_k. \tag{35}$$

The D-matrix is block-diagonal in n (same principal quantum number shell) and diagonal in m (selection rule from axial symmetry).

c. Two-electron integrals. The two-electron repulsion integrals are computed between MOs projected onto atom-centered functions via the prolate spheroidal wavefunctions. For each MO k, the projection onto center $\alpha \in \{A, B\}$ gives expansion coefficients $c^{(n,l,m)}_{k,\alpha}$, and the two-electron integrals are

$$\langle kl|rs\rangle = \sum_\alpha \sum_{n_i l_i m_i} c^*_{k,\alpha} c_{l,\alpha} c^*_{r,\alpha} c_{s,\alpha}\, F^0(\alpha; n_k l_k, n_r l_r), \tag{36}$$

where F^0 are the same-center Slater direct integrals. Cross-atom two-electron integrals are small at equilibrium geometry and can be included perturbatively.

C. CI Expansion

The CI expansion is over Slater determinants of the molecular Sturmian spin-orbitals. The Hamiltonian assembly follows the same Slater–Condon rules as the atom-centered FCI (diagonal, single excitation, double excitation), with matrix elements computed from Eqs. (34)–(36).

D. Self-Consistency

The self-consistency loop is:

1. Initialize $p_0^{(0)} = \sqrt{(Z_A^2 + Z_B^2)/2}$.
2. At each iteration k:
   (a) Solve the prolate spheroidal eigenvalue problem for all $\beta_k^{(k)}$ at the current $p_0^{(k)}$.
   (b) Build the FCI Hamiltonian with the current $p_0^{(k)}$, $\beta_k^{(k)}$, and $\gamma^{(k)}$.
   (c) Diagonalize to obtain $E_{\text{mol}}^{(k)}$.
   (d) Update $p_0^{(k+1)} = \sqrt{-2E_{\text{mol}}^{(k)}}$ (with damping $\alpha \leq 0.5$ if needed).
3. Converge when $|p_0^{(k+1)} - p_0^{(k)}| < \varepsilon$.

XIII. POLYATOMIC EXTENSION

The bond sphere is the natural building block for polyatomic molecules. A general molecule is conjectured to be a *graph of bond spheres*: each atom is a vertex (pole), each bond is an edge carrying a bond sphere S^3 parameterized by γ_{ij}, and bond spheres share poles at atomic vertices.

a. Water (H_2O). Three vertices (O, H, H), two bond edges (O–H, O–H). The H–O–H bond angle enters as a relative rotation between bond sphere orientations at the shared oxygen pole.

b. Status. The polyatomic construction is conjectural beyond the diatomic case. The main open question is how the shared pole mediates interactions between adjacent bond spheres.

XIV. WHAT THIS PAPER DOES NOT PROVE

1. **Convergence rate** of the self-consistency iteration is asserted to exist but not proven to be rapid.
2. **Two-electron integrals** on the bond sphere in the Sturmian basis require $SO(4)$ Clebsch–Gordan coefficients not yet derived in GeoVac notation.
3. **Polyatomic extension**: inter-bond coupling at shared atomic poles is not yet worked out.
4. **Non-compact generators** of $SO(4,2)$ may affect the Sturmian condition in the two-pole geometry.
5. **Inter-shell coupling** ($n' \neq n$): the D-matrix is block-diagonal in n. Cross-n coupling would require the full conformal group $SO(4,2)$.

XV. CONCLUSION

A diatomic molecule is a single S^3 with two weighted poles—the bond sphere. The internuclear distance encodes as a polar angle γ, and all cross-atom matrix elements are $SO(4)$ Wigner D-matrix elements evaluated at the bond rotation $e^{i\gamma A_y}$. The natural basis for this geometry is the Sturmian basis, in which all functions share a single p_0, all live on the same S^3, and the Shibuya–Wulfman cross-center integrals are exact D-matrix elements (Eq. (23)).

The β_k eigenvalues of the molecular Sturmian functions are determined by the prolate spheroidal eigenvalue problem—whose coordinate surfaces are the confocal ellipsoids of the bond sphere metric. The uniform diagonal theorem (Theorem 3) establishes that the one-electron diagonal is $-p_0^2/2$ for every orbital, with all nuclear information encoded in the off-diagonal D-matrix structure.

Every atom-centered approximation to this basis fails for heteronuclear molecules (Theorem 2): a shared p_0 determined by the deeper well pushes the lighter center's orbitals into positive energy, and no atom-centered cross-nuclear correction can compensate because the atom-centered function has the wrong spatial extent. The molecular Sturmian—the prolate spheroidal eigenfunction—has the correct extent near both centers and achieves the binding threshold.

The single-atom lattice of Papers 0–7 is the $\gamma = 0$ limit. The 18 symbolic proofs, the dimensionless vacuum principle, and the universal constant $-1/16$ are unchanged. The Runge–Lenz alignment argument explains why s–p hybridization is a dynamical consequence of the two-center potential, not an imposed ansatz. For heteronuclear molecules, the bond sphere becomes an egg—a prolate spheroid whose asymmetry encodes bond polarity.

Connection to Paper 0: the quantum numbers (n, l, m) that label nodes on the S^3 lattice are the same quantum numbers that arise from the geometric packing problem of Paper 0. The bond sphere extends this packing to two centers, with the D-matrix providing the natural inter-center coupling that respects the $SO(4)$ structure established by the packing geometry. The Sturmian basis is the bridge between the discrete graph topology and the continuous molecular wavefunctions: it lives on the graph's S^3 by construction and provides the correct two-center eigenfunctions through prolate spheroidal separation.

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